Statement of Purpose

Md Toki Tahmid

I am passionate about advancing research in computational biology, particularly by developing computational frameworks to address foundational challenges in understanding biological processes and disease progression. During my undergraduate studies, I spent over three years working in this domain, including collaborations with MIT's Computational Connectomics Lab and CMU's XuLAB. My work spans genome language modeling, computational proteomics, and biomedical informatics. These experiences, along with a strong research record, publications, and academic achievements, have solidified my commitment to pursuing a Ph.D. in Princeton University's Computer Science program. This program's unparalleled interdisciplinary research environment will enable me to bridge critical gaps in biological understanding and contribute to improving human health outcomes.

Research Experience:

Building on my early fascination with biological interactions, I pursued computational proteomics under Professor Md. Shamsuzzoha Bayzid. To address class imbalance in protein-nucleic acid docking prediction, I developed TransBind [1], using class-weighted training with protein language models (PLMs). This work is **accepted in Nature Communications Biology** as a first-author paper. Recognizing the synergy between structural insights and PLMs, I later developed EmbedSimScore [2] (First author), combining local substructures and global representations for protein similarity beyond TM-Score. **Accepted at SSL - Theory and Practice, NeurIPS 2024**, it offers a holistic protein similarity metric. While working on protein language models, I identified a key limitation of biological LLMs: their fixed context windows, which require longer sequences to be segmented, leading to information loss. To address this, I collaborated with Professor Yue Dong at UCR to develop BiRNA-BERT [3], an RNA language model capable of processing arbitrarily long sequences. Using an adaptive tokenization scheme with Byte Pair Encoding (BPE) and dual pre-training for sequence- and token-level tasks, BiRNA-BERT eliminates the need for segmentation, extending the context window sixfold. **Recognized as a spotlight paper at the NeurIPS 2024 ENLSP workshop** (first author), it achieves state-of-the-art performance on long-sequence RNA tasks, outperforming models 27 times more computationally intensive. BiRNA-BERT is currently under revision in Nature Communications Biology.

My fascination with structural and functional biology at the cellular level inspired me to explore these processes at the microscopic level. During my internship at XuLab, I worked with Professor Min Xu on developing TomoPicker [4](**Accepted at the NeurIPS Workshop on MLSB 2024**), a positive-unlabeled learning-based approach for efficient particle picking in cellular cryo-electron tomography (cryo-ET) datasets. Particularly, my work on a tracing algorithm that can seamlessly convert the 2D annotations made on each cryo-ET slice into a volumetric annotation is used in the lab's other projects while validating the performance of 3D annotation methods. My interest in computational biology naturally extended to computational neuroscience, inspired by the parallels between biological and artificial neural networks. At MIT's Computational Connectomics Lab, I collaborated with Professors Nir Shavit and Lu Mi to study the brain's connectome—modeled as graphs of neurons and synapses—and its functional underpinnings. Building on this, I co-developed StructureReps [5] (**First author, accepted at NeuREPS workshop, 2024**), an architecture-agnostic framework that encodes both biological and artificial neural networks as graphs using graph representation learning.

While computational biology is now my primary focus, my earlier experiences applying machine learning to diverse challenges helped shape this direction. During my undergraduate studies, I developed MD-CardioNet [6], a deep-learning model for arrhythmia detection, published in the **IEEE Journal of Biomedical and Health Informatics**. Additionally, during the COVID-19 outbreak, I worked on multi-modal epidemic forecasting, resulting in a publication [7] in **Results in Physics** (75 citations). My interdisciplinary perspective on machine learning, natural sciences, and real-life applications has led to several other works [8–15], which are currently under review in prominent journals and conferences.

Future Research Goals:

From my diverse research experiences in computational biology and health informatics, I have developed a strong conviction that computational approaches can reveal complex patterns in biological processes and disease mechanisms. The 2024 Nobel Prize in Chemistry for AI-powered proteomics

solidified my belief that the next transformative AI breakthroughs will be in computational healthcare for which understanding the critical biological interactions within the human body from a multi-dimensional computational perspective is very crucial.

To this end, I would love to participate in the development of multimodal reasoning models (AI models, including LLMs) capable of integrating diverse biological and clinical data modalities: genetic sequences, expression profiles, clinical vitals, pathological reports from both medical imaging and metadata, and the extensive body of clinical literature. ***Professor Mona Singh's*** research on analyzing high-throughput datasets to infer protein interactions, predict mutation impacts in the context of disease, and analyze biological networks aligns closely with my previous research works and goal of integrating diverse biological data to understand disease mechanisms. I believe integrating multi-modal biological information and pathways into the AI based reasoning models will present a more holistic and explainable view of biological processes, which I would be delighted to explore with professor Mona Singh. ***Professor Olga Troyanskaya's*** works on large-scale biological data sets, particularly the focus of her lab in algorithms for integration of data from multiple data sources aligns perfectly with my previous experience in biological foundation models, computational genomics and also aligns perfectly with my future research goals. To this end, I would also love to work with ***professor Yuri Pritykin***, whose primary research agenda lies in multi-modal single-cell, spatial and genome editing technologies to address fundamental problems in biology and medicine.

Upon completing my Ph.D., I aim to pursue an academic or research-driven role where I can continue to develop computational frameworks to address foundational questions in biology and disease mechanism. Princeton's Department of Computer Science, with its emphasis on collaborative innovation as demonstrated by the works of Professors Mona Singh, Olga Troyanskaya, and Yuri Pritykin aligns perfectly with my immediate and long-term goals. Its collaborative culture and access to world-class resources provide the ideal foundation for achieving my vision of advancing computational tools to unravel the complexities of biological systems and disease mechanisms.

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